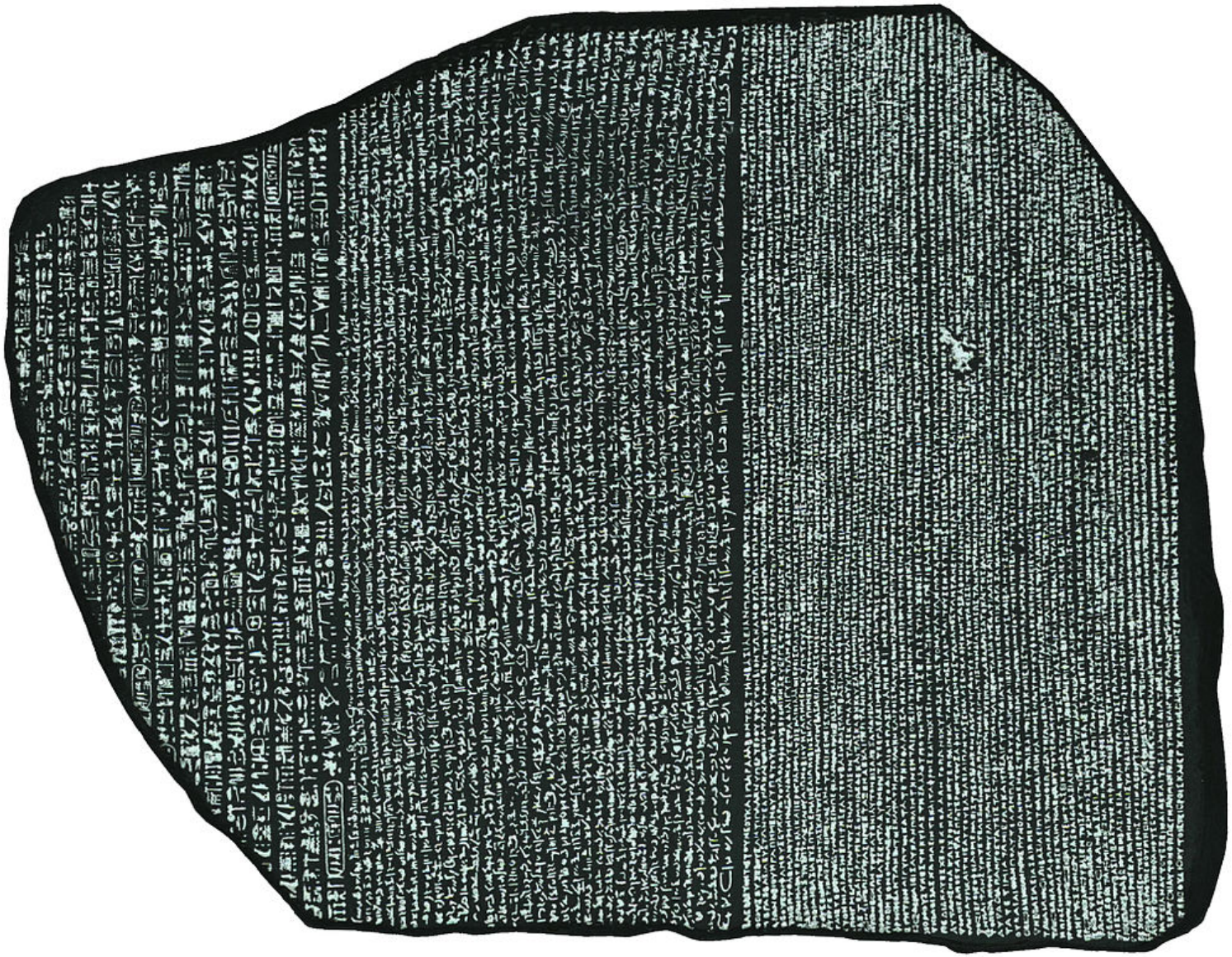
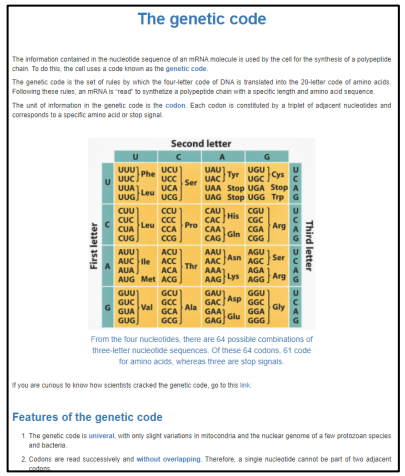


The genetic code

The effect of mutations

Supporting material



The genetic code

The information contained in the nucleotide sequence of an mRNA molecule is used by the cell for the synthesis of a polypeptide chain. To do this, the cell uses a code known as the genetic code.

The genetic code is the set of rules by which the four-letter code of DNA is translated into the 20-letter code of amino acids. Following these rules, an mRNA is "read" to synthetize a polypeptide chain with a specific length and amino acid sequence.

The unit of information in the genetic code is the codon. Each codon is constituted by a triplet of adjacent nucleotides and corresponds to a specific amino acid or stop signal.

Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC: Phe; UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr; UAA, UAG: Stop	UGU, UGC: Cys; UGA: Stop; UGG: Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His; CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile; AUG: Met	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn; AAA, AAG: Lys	AGU, AGC: Ser; AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp; GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

From the four nucleotides, there are 64 possible combinations of three-letter nucleotide sequences. Of these 64 codons, 61 code for amino acids, whereas three are stop signals.

If you are curious to know how scientists cracked the genetic code, go to this link.

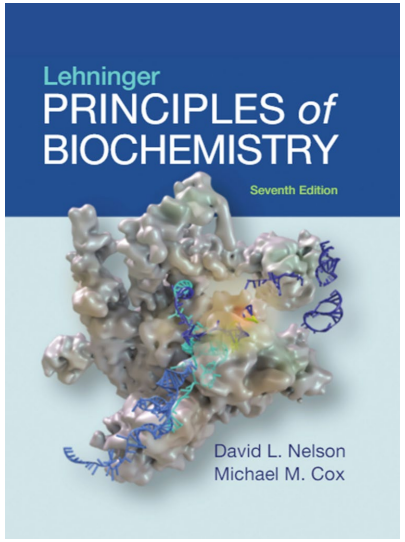
Features of the genetic code

1. The genetic code is universal, with only slight variations in mitocondria and the nuclear genome of a few protozoan species and bacteria.
2. Codons are read successively and without overlapping. Therefore, a single nucleotide cannot be part of two adjacent codons.

Study material

ADI document:

The genetic code



Reading

Lehninger's PoB Chapter 27:

27.1: The genetic code

Learning goals

- Characteristics of the genetic code
- Codon and anticodon interaction
- Types of mutations and their impact on the polypeptide chain

The genetic code

Deciphering the genetic code was one of the greatest scientific developments of the 20th century

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	A
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G



Severo Ochoa



Marshall Nirenberg



Har Gobind Khorana



Robert Holley



Crick's adaptor hypothesis

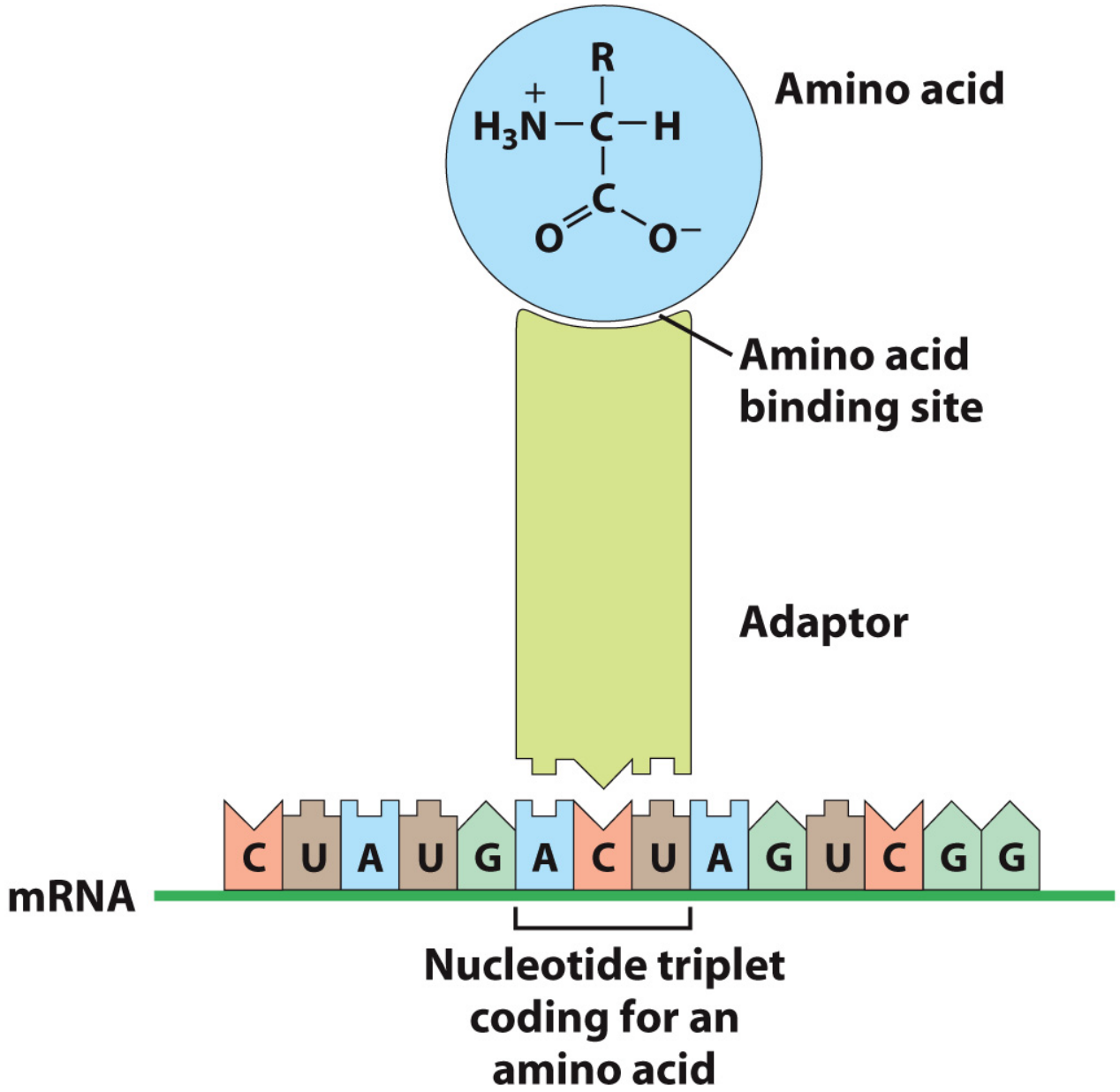


Figure 27-2
Lehninger Principles of Biochemistry, Seventh Edition
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The genetic code

First position (5′ end)	Second position				Third position (3′ end)
	U	**C**	**A**	**G**	
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	**U**
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	**C**
	UUA Leu	UCA Ser	UAA Stop	UGA Stop	**A**
	UUG Leu	UCG Ser	UAG Stop	UGG Trp	**G**
C	CUU Leu	CCU Pro	CAU His	CGU Arg	**U**
	CUC Leu	CCC Pro	CAC His	CGC Arg	**C**
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	**A**
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	**G**
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	**U**
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	**C**
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	**A**
	AUG Met[b]	ACG Thr	AAG Lys	AGG Arg	**G**
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	**U**
	GUC Val	GCC Ala	GAC Asp	GGC Gly	**C**
	GUA Val	GCA Ala	GAA Glu	GGA Gly	**A**
	GUG Val	GCG Ala	GAG Glu	GGG Gly	**G**

Voet 3rd edition

Features of the genetic code

1- Nearly universal

2- Based on triplets of nucleotides named codons

3- Continuous: codons are read consecutively, without gaps or overlap

4- Unambiguous: each codon specifies a single amino acid or a stop signal

5- Degenerate (redundant): a single amino acid can have multiple codons (synonymous codons)

6- Redundancy often occurs at the third position of the codon

7- Codons differing by a single nucleotide often encode amino acids with similar chemical properties

8- AUG is the usual initiation codon, which encodes methionine

9- There are three nonsense codons (termination or stop codons): UAA, UAG and UGA

Molecular recognition of codon by tRNA

- The codon sequence is complementary to the anticodon sequence
- The codon in mRNA pairs with the anticodon in tRNA via hydrogen bonding
- The alignment is antiparallel

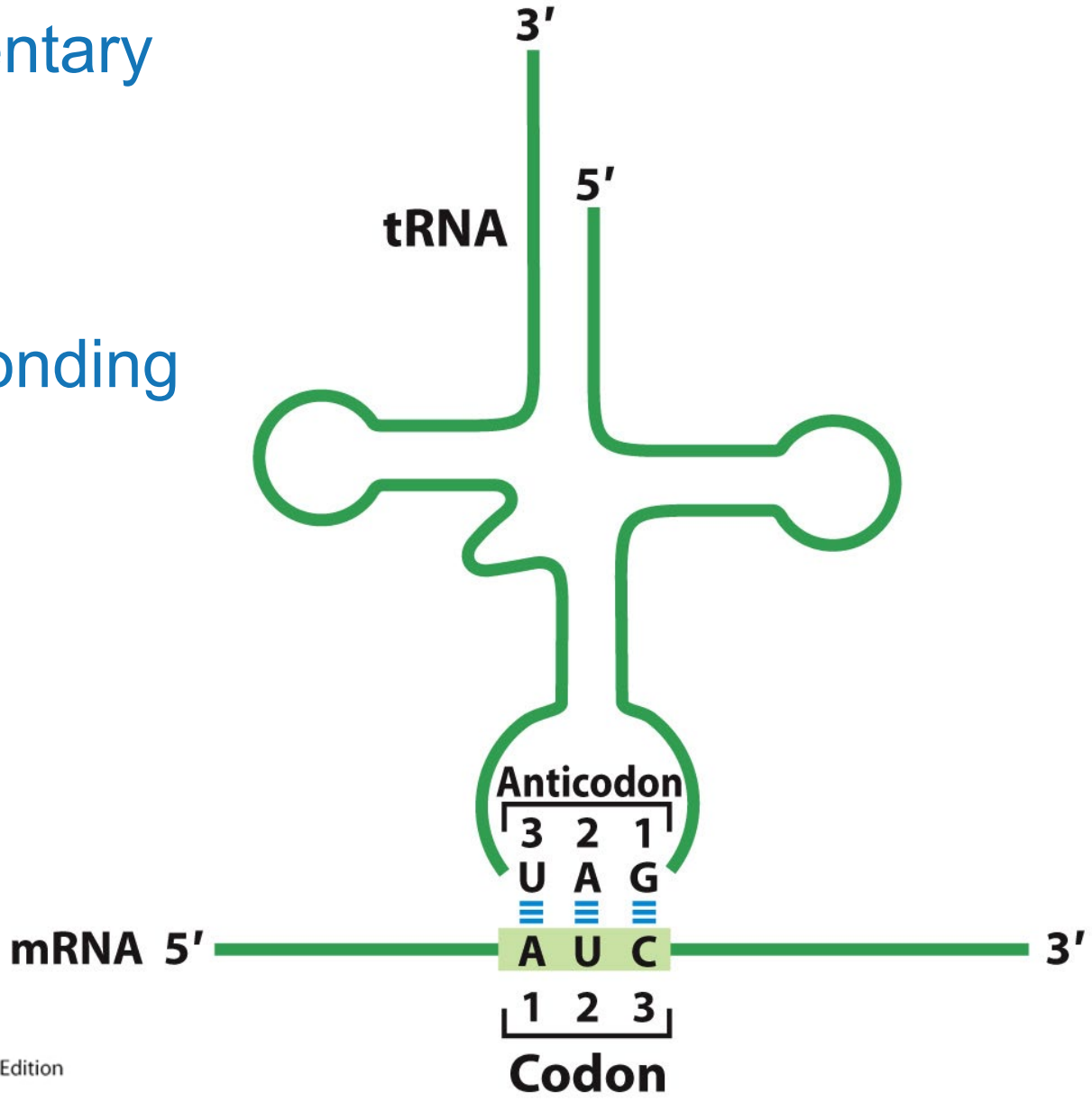


Figure 27-8a
Lehninger Principles of Biochemistry, Seventh Edition
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The effects of mutations

… UCC GGA UAU GUU CAA GCU AGA … Wild-type mRNA

… Ser Gly Tyr Val Gln Ala Arg … Wild-type polypeptide

… UCC GGA UAU GUU CAA G**G**U AGA … Missense mutation

… Ser Gly Tyr Val Gln **Gly** Arg …

… UCC GGA UAU GUU CAA GC**G** AGA … Synonymous (silent) mutation

… Ser Gly Tyr Val Gln **Ala** Arg …

… UCC GGA UA**A** GUU CAA GCU AGA … Nonsense mutation

… Ser Gly **Ter**

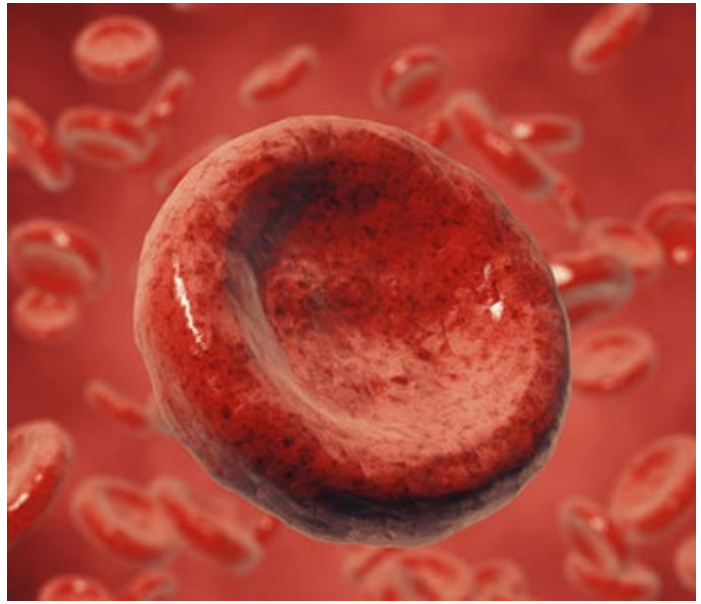
… UCC GGA UA**C** UGU UCA AGC UAG … Frameshift mutation

… Ser Gly **Tyr** **Cys** **Ser** **Ser** **Ter**

… UCC GGA UAU **AUC** GUU CAA GCU … In-frame insertion

… Ser Gly Tyr **Ile** **Val** **Gln** **Ala** …

Mutations in hemoglobin genes



1. HbS (Sickle cell anemia)

   β^{6} GAG → GUG

   Glu Val

2. Hb McKees Rocks

   β^{144} AAGUAUCACUAA → AAGUAACACUAA

   Tyr Stop codon

3. Hb Constant Spring

   α^{141} CGUUAA → CGUCAAGCU----------UAA

   Stop codon Gln α172

Open questions for discussion

- Why has the genetic code remained so highly conserved throughout evolution?
- Why is redundancy often found at the third position of the codon?
- What are the advantages of a degenerate genetic code?
- Why do some similar amino acids have similar codons?

